

PROMPT ENGINEERING

1. Objective

The objective of this assignment was to understand and apply prompt engineering techniques to a hospitality use case. Five prompts were designed using different approaches to improve the accuracy, consistency, and usefulness of Large Language Model (LLM) responses.

The prompts demonstrate the use of **system instructions, user instructions, context framing, few-shot prompting, and structured output schemas**.

2. Introduction

Prompt engineering is the process of designing effective instructions for an AI model to generate a desired response. A well-structured prompt clearly defines the task, provides relevant context, establishes constraints, and specifies the expected format.

In the hospitality domain, prompt engineering can be used for tasks such as handling guest requests, classifying complaints, answering hotel-related questions, routing requests to the appropriate department, and generating professional customer-service responses.

For this assignment, a hotel guest-support scenario was selected and five different prompts were created.

3. Prompt 1 – Guest Request Classification

Purpose

This prompt is designed to classify incoming hotel guest messages into predefined categories. The system instruction defines the role of the AI and the available categories, while the user provides the guest message.

Prompt

SYSTEM:

You are a professional hotel guest-support assistant. Your task is to classify

incoming guest messages accurately.

Classify every message into exactly one of these categories:

- ROOM_SERVICE
- HOUSEKEEPING
- MAINTENANCE
- FRONT_DESK
- BILLING
- GENERAL

Do not create new categories. Consider the main intent of the guest's message.

Return the result only in the requested format.

USER:

Classify the following guest message:

"My air conditioner is not cooling the room and it is becoming uncomfortable."

Return the result in this format:

```
{  
  "category": "<category>",  
  "reason": "<short explanation>",  
  "confidence": <number between 0 and 1>  
}
```

OUTPUT

```
{  
  
  "category": "MAINTENANCE",  
  
  "reason": "The air conditioner is not functioning properly and requires repair.",  
  
  "confidence": 0.99
```

```
}
```

This prompt demonstrates **system instructions and user instructions**. The system establishes the rules and classification categories, while the user provides the actual task.

4. Prompt 2 – Context Framing

Purpose

This prompt demonstrates how providing relevant context can help the AI generate a response based on specific hotel information rather than relying on general assumptions.

Prompt

SYSTEM:

You are a professional hotel customer-service assistant. Respond politely, clearly, and empathetically to hotel guests.

Do not promise services that are not available. If the guest's request requires hotel staff intervention, clearly state that the request needs to be handled by the appropriate hotel team.

USER:

Hotel Context:

Hotel Name: Grand Horizon Hotel
Check-in Time: 2:00 PM
Check-out Time: 12:00 PM
Breakfast Hours: 7:00 AM to 10:30 AM
24-hour Front Desk: Yes
Room Service: Available until 11:00 PM

Guest Message:

"I have an early flight tomorrow. Can I request a breakfast box at 5:30 AM?"

Respond based only on the information provided in the hotel context.

OUTPUT

I'm sorry, but the hotel context only confirms that breakfast is served from 7:00 AM to 10:30 AM. It does not provide information about breakfast boxes or whether an early breakfast can be arranged.

This request would need to be handled by the **appropriate hotel team/front desk**.

This prompt demonstrates **context framing**. Relevant hotel information is explicitly provided so that the AI can generate a response appropriate to the specific hotel environment.

5. Prompt 3 – Few-Shot Prompting

Purpose

This prompt demonstrates few-shot prompting, where examples are provided to show the model the expected response style before it receives a new guest request.

Prompt

SYSTEM:

You are a hotel guest-support assistant. Generate short, polite, and professional responses to guest requests.

Follow the style demonstrated in the examples.

EXAMPLE 1:

Guest:

"The Wi-Fi isn't working in my room."

Response:

"We're sorry for the inconvenience. Our team will be happy to assist you with the Wi-Fi issue. Please contact the front desk so we can help resolve it as quickly as possible."

EXAMPLE 2:

Guest:

"Can I get two extra towels?"

Response:

"Certainly. We can arrange two additional towels for your room. Our housekeeping team will assist with the request shortly."

EXAMPLE 3:

Guest:

"What time does breakfast start?"

Response:

"Breakfast is served from 7:00 AM to 10:30 AM. We hope you enjoy your breakfast with us."

USER:

Guest:

"The room next to mine is very noisy. Could someone please help?"

OUTPUT

We're sorry for the inconvenience. Our team will be happy to assist with the noise issue. Please contact the front desk so we can help resolve it as quickly as possible.

Few-shot prompting provides examples of previous inputs and responses. This helps establish the desired **tone, structure, and style** for handling new guest requests.

6. Prompt 4 – Structured Output Schema

Purpose

This prompt demonstrates the use of a structured output format. A defined JSON structure makes the AI response consistent and easier for an application to process.

Prompt

SYSTEM:

You are a hotel guest-support assistant responsible for analyzing guest requests.

Extract the important information from the guest message and return the result only as valid JSON.

The output must contain:

- intent
- urgency
- department
- guest_request
- recommended_action

Allowed urgency values:
LOW, MEDIUM, HIGH

USER:

Guest Message:

"I've been waiting for room service for more than an hour. I ordered dinner at 8 PM and still haven't received it."

Return the output using this structure:

```
{
  "intent": "",
  "urgency": "",
  "department": "",
  "guest_request": "",
  "recommended_action": ""
}
```

OUTPUT

```
{
  "intent": "Room service delay",
  "urgency": "MEDIUM",
  "department": "ROOM_SERVICE",
  "guest_request": "The guest wants an update and resolution for the
  delayed dinner order.",
  "recommended_action": "Contact the room service team to check the
  order status and expedite delivery."
}
```

The structured schema ensures that the AI consistently returns information in a predefined format. This is particularly useful when AI responses need to be consumed by software systems.

7. Prompt 5 – Combined Prompt Engineering Techniques

Purpose

The fifth prompt combines multiple prompt engineering techniques into a single hospitality use case.

It uses **system instructions, context framing, few-shot examples, user instructions, and output constraints.**

Prompt

SYSTEM:

You are an AI assistant for hotel guest support.

Your responsibilities are to:

1. Understand the guest's request.
2. Identify the main issue.
3. Determine the appropriate hotel department.
4. Determine the urgency.
5. Provide a polite and professional response.

Use only the information provided in the hotel context.
Do not invent hotel policies or services.

HOTEL CONTEXT:

Hotel Name: Grand Horizon Hotel

Available Departments:

- Front Desk
- Housekeeping
- Maintenance
- Room Service
- Billing

Hotel Policies:

- Check-in: 2:00 PM
- Check-out: 12:00 PM
- Breakfast: 7:00 AM to 10:30 AM
- Front Desk: Available 24 hours

FEW-SHOT EXAMPLES:

Guest:

"The bathroom light is not working."

Department: Maintenance

Urgency: MEDIUM

Guest:

"I need an extra pillow."

Department: Housekeeping

Urgency: LOW

USER:

Guest:

"There is water leaking from the ceiling in my room and the floor is becoming wet."

Return only the following JSON structure:

```
{
  "issue": "",
  "department": "",
  "urgency": "",
  "response": ""
}
```

OUTPUT

```
{
  "issue": "Water leaking from the ceiling and making the floor wet",
  "department": "Maintenance",
  "urgency": "HIGH",
  "response": "We're sorry for the inconvenience. This requires immediate attention from our maintenance team. Please contact the front desk so the issue can be addressed as quickly as possible."
}
```

This prompt combines multiple techniques to provide the model with clear instructions, relevant context, examples, and a defined output format.

8. Key Learnings

The exercise demonstrated that prompt structure has a significant effect on the quality and consistency of AI-generated responses.

System instructions establish the role and behavior of the model. **Context framing** provides the information required to handle a specific situation. **Few-shot prompting** helps guide the model using examples, while **structured output** ensures that responses follow a predefined format.

Combining these techniques can make AI systems more reliable and suitable for real-world applications such as hospitality customer support.

9. Conclusion

The assignment demonstrated five practical approaches to prompt engineering using a hospitality scenario. The prompts covered **system and user instructions, context framing, few-shot prompting, structured output schemas, and combined prompting techniques**.

These techniques can be applied to hospitality applications for handling guest requests, classifying issues, generating customer-service responses, and extracting structured information. The exercise provided practical understanding of how carefully designed prompts can improve the consistency, relevance, and usability of LLM responses.

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